Simulation of a large **wildfire** in a coupled fire-atmosphere **model**

JB Filippi, F Bosseur, C Mari, C Lac - Atmosphere, 2018 - mdpi.com

… fire **spread model** is proposed for predicting fine- to large-scale effects of this extreme **wildfire**, showing that such **simulation** is … (see [31] for a **review** of numerical interface tracking and …

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Integrated national-scale assessment of **wildfire** risk to human and ecological values

MP Thompson, DE Calkin, MA Finney, AA Ager… - … Research and Risk …, 2011 - Springer

… **wildfire** entailed use of FSim, a **wildfire simulation** system. The specifics of FSim, including details on how fire **behavior** … impacted by fire) through **review** of many recovery plans and …

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Wildfire risk in the wildland–urban interface: a **simulation** study in northwestern Wisconsin

AB Massada, VC Radeloff, SI Stewart… - … and **Management**, 2009 - Elsevier

… We simulated **wildfire spread** and **behavior** with two commonly used models: FARSITE (Finney, 1998) and MTT (Finney, 2002), as implemented in FlamMap (Finney, 2006). FARSITE is …

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Homeowners willingness to pay to reduce **wildfire** risk in wildland urban interface areas: Implications for targeting financial incentives

JJ Sanchez, TP Holmes, J Loomis… - International Journal of …, 2022 - Elsevier

… This suggests the importance of risk perceptions in adoption of risk mitigation **behavior**. … **wildfire** risk perceptions and mitigation **behavior**, not explicitly revealed in the basic MNL **model**, …

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Socio-Ecological Perceptions of **Wildfire Management** and Effects in Greece. Fire 2021, 4, 18

P Palaiologou, K Kalabokidis, A Troumbis, MA Day… - 2021 - academia.edu

… emphasis on reforming **wildfire management** policies to deal … of climate change in **wildfire** frequency and **behavior**. In addition… -based **survey** of professionals in **wildfire management** and …

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An integrative **review** of empirical research on perceptions and behaviors related to prescribed burning and **wildfire** in the United States

LN Dupéy, JW Smith - Environmental **management**, 2018 - Springer

… (evacuation), also increased the likelihood of engaging in mitigation **behavior**. These two studies highlight how emerging research is beginning to explore a wider collection of …

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Model comparisons for estimating carbon emissions from North American **wildland fire**

NHF French, WJ De Groot, LK Jenkins… - Journal of …, 2011 - Wiley Online Library

… by the sixth **model** used in this study, the Canadian **Forest Fire Behavior Prediction** (FBP) … Here we focus our **review** on total carbon emissions estimates from burning, with the aim of …

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The role of UAV-IoT networks in future **wildfire** detection

OM Bushnaq, A Chaaban… - IEEE Internet of Things …, 2021 - ieeexplore.ieee.org

… In [17], a **survey** on the flying ad hoc networks is presented. The … , we assume a simplistic fire **spread model**, wherein fire evolves … In Section II, we present the system **model** and the main …

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Exploring adoption of the **wildland fire** decision support system: End user perspectives

P Noble, TB Paveglio - Journal of Forestry, 2020 - academic.oup.com

… WFDSS is a web-based application that synthesizes geospatial data, fire **behavior model** outputs, risk analysis frameworks, and cost evaluation tools. It is intended to help fire managers …

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Social amplification of **wildfire** risk: the role of social interactions and information sources

H Brenkert-Smith, KL Dickinson, PA Champ… - Risk …, 2013 - Wiley Online Library

… , we develop a conceptual **model** of **wildfire** risk perceptions that … To the extent that people are aware of these **dynamics**, we … Here, we briefly describe the **survey** design, as well as the …

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Psychological factors and social processes influencing **wildfire** smoke protective **behavior**: Insights from a case study in Northern California

FN Santana, DJX Gonzalez, G Wong-Parodi - Climate Risk **Management**, 2021 - Elsevier

… health **behavior**. Finally, we propose a conceptual framework of **wildfire** smoke protective **behavior**. … risk and protective health literatures related to **wildfire** smoke response. Our findings …

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Data assimilation for wildland fires

J Mandel, JD Beezley, JL Coen… - IEEE Control Systems …, 2009 - ieeexplore.ieee.org

… A **wildland fire** interacts with the atmosphere **dynamics** through fluxes of momentum, water … Then the state of the **model** is the probability distribution of possible **wildfire** scenarios. The …

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Simulation of **forest fire** fronts using cellular automata

AH Encinas, LH Encinas, SH White, AM Del Rey… - … in Engineering Software, 2007 - Elsevier

… Specifically, the **model** proposed in this work introduces a more accurate factor of **propagation** from a diagonal neighbor cell and includes, in a detailed form, the rate of fire **spread**. …

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Spatiotemporal **dynamics** of simulated **wildfire**, forest **management**, and forest succession in central Oregon, USA

AMG Barros, AA Ager, MA Day, HK Preisler, TA Spies… - Ecology and …, 2017 - JSTOR

… **simulation model** Envision to analyze long-term **wildfire dynamics** and the effects of different fuel **management** … We simulated a 50-year future where fuel **management** activities were …

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Forest fire spread model based on the grey system theory.

C Lv, J Wang, F Zhang - Journal of Supercomputing, 2020 - search.ebscohost.com

… **forest fire spread model** developing. By preprocessing and fusing the MODIS remote sensing data, the sequence data of the grey **model** … algorithm, **forest fire spread** GM(1, 1) **model** was …

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Comparing production function models for **wildfire** risk analysis in the wildland–urban interface

DE Mercer, JP Prestemon - Forest policy and economics, 2005 - Elsevier

… to **wildfire** … **wildland fire management** decisions in the wildland–urban interface. Our results also emphasize the importance of including such variables in statistical models of **wildfire** risk …

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Carbon **dynamics** of North American boreal forest after stand replacing **wildfire** and clearcut logging

M Seedre, BM Shrestha, HYH Chen… - Journal of forest …, 2011 - Taylor & Francis

… this **review** was to synthesize the literature on C **dynamics** of … disturbances, stand replacing **wildfire** and clearcut logging. … of **wildfire** and harvesting stands with different **management** …

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Conflict and collaboration in **wildfire management**: the role of mission alignment

CJ Fleming, EB McCartha… - … Administration **Review**, 2015 - Wiley Online Library

… By understanding where differences can arise and whether they are managed well, we provide insight into **dynamics** of relationships that will need to be examined and addressed. …

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Health effects of **wildfire** smoke exposure

CF Gould, S Heft-Neal, M Johnson… - Annual **Review** of …, 2024 - annualreviews.org

… and social systems adapt to new **dynamics** (eg, increased **wildfire** activity) by changing aspects of … smoke days have low to medium PM2.5 concentrations, using the quartic **model** we …

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Toward improved communication about **wildland fire**: mental models research to identify information needs for natural resource **management**

M Zaksek, JL Arvai - Risk Analysis: An International Journal, 2004 - Wiley Online Library

… **model** was created based on an extensive **review** of the literature and consultations with several **wildland fire** … as factors that influence **wildland fire** ignition and **spread**. Similarly, only …

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